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Assessment of deep learning models integrated with weather and environmental variables for wildfire spread prediction and a case study of the 2023 Maui fires

Jiyeon Kim · Yingjie Hu · Negar Elhami-Khorasani · Kai Sun · Ryan Zhenqi Zhou

Environmental Science, Computer Science · 5 September 2025

TLDR This work assesses the ability of five typical deep learning models integrated with weather and environmental variables for **wildfire spread prediction** based on over ten years of **wildfire** data in the state of Hawaii and identifies important weather and environmental factors associated with the 2023 Maui **wildfires**. Expand

[PDF] arXiv Save Cite

Wildfire Spread Prediction using Machine Learning Algorithms

Aman Singh · Rakesh Yadav · Gadug Sudhamshu · Aryan Basnet · Rahmat Ali

Environmental Science, Computer Science · International Conference on Computing… · 6 July 2023

TLDR This study utilized machine learning algorithms, including Decision Tree Regression, XG Boostregression, and Artificial Neural Networks, to predict the **spread** of **wildfires** using the Next Day **Wildfire** dataset, which includes satellite images, weather, and geography conditions aggregated across the United States from 2012 to 2020. Expand

IEEE Save Cite

An Enhanced Wildfire Spread Prediction Using Multimodal Satellite Imagery and Deep Learning Models

R. Zakari · O. Malik · Ong Wee-Hong · Environmental Science, Computer Science · Remote Sensing Applications Society and… · 1 June 2025

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Evolutionary Statistical System Based on Novelty Search: A Parallel Metaheuristic for Uncertainty Reduction Applied to Wildfire Spread Prediction

Jan Strappa · Paola Caymes-Scutari · G. Bianchini · Environmental Science, Computer Science · Algorithms · 15 December 2022

TLDR The method, Evolutionary Statistical System based on Novelty Search (ESS-NS), outperforms the quality obtained by its competitors in the authors' experiments and execution times are faster than other methods for almost all cases. Expand

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Wildfire Spread Prediction Model Calibration Using Metaheuristic Algorithms

Jorge Pereira · Jérôme Mendes · J. S. Junior · C. Viegas · J. Paulo

Environmental Science, Engineering · Annual Conference of the IEEE Industrial… · 17 October 2022

TLDR This paper uses 3 metaheuristic algorithms, genetic algorithm (GA), differential evolution (DE) and simulated annealing (SA), for calibration of input parameters from the Rothermel model, and shows that these algorithms provide a precise **wildfire spread prediction** accounting for the uncertainties in the model’s selected parameters. Expand

1 PDF IEEE Save Cite

Crowd-Sourced Wildfire Spread Prediction With Remote Georeferencing Using Smartphones

Nikos Bogdos · E. Manolakos · Environmental Science, Computer Science · IEEE Access · 2019

TLDR CITISENS offers the following unique advantages: real-time **prediction** of burn probabilities, dynamic assimilation of citizen-reported hotspots into ongoing simulations for improved predictive accuracy, and decision support to issue citizen alarms based on the estimated time-dependent risk at their location due to an approaching **wildfire**. Expand

3 PDF IEEE Save Cite

Wildfire Spread Prediction and Assimilation for FARSITE Using Ensemble Kalman Filtering

Thayjes Srivas T. Artés R. A. Callafon I. Altintas Computer Science, Environmental Science ·

International Conference on Conceptual Structures · 1 June 2016

43 Publisher Save Cite

Exploring Directions for Establishing Forest Fire Containment Lines through Forest Fire Spread Prediction Simulations: A Case Study of 2022 Uljin-Samcheok Wildfire

Nam Gyun Kim Chan-ho Yeom Chun-Geun Kwon Environmental Science ·

Crisis and Emergency Management: Theory and… · 31 May 2024

In this study, the extent of forest fire damage over time was predicted through numerical simulations for the 2022 Uljin-Samcheok forest fire. Also, the area of suppressed forest fire **spread** through… Expand

Publisher Save Cite

Shapley-based interpretation of deep learning models for wildfire spread rate prediction

Faiza Qayyum N. A. Samee Maali Alabdulhafith Ahmed Aziz Mohammad Hijjawi

Fire Ecology · 25 January 2024

Antecedentes Predecir la propagación de fuegos de vegetación, es vital para poder determinar sus efectos detrimentales. Mientras que durante muchos años numerosos estudios han profundizado en… Expand

8 PDF Springer Nature Save Cite

California Wildfire Spread Prediction using FARSITE and the Comparison with the Actual Wildfire Maps using Statistical Methods

Yujia Hao Environmental Science, Computer Science · 2018

TLDR The results show that the performance of FARSITE simulation model is acceptable, which can be an option for predicting future **wildfires**, and statistical methods were applied to measure the similarity between the predictive **wildfires** area and the actual **wildfire** areas. Expand

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